

Portfolio Risk Measurement and VaR Analysis

Executive Summary

This project presents a comprehensive quantitative risk analysis of a Swiss equity portfolio using advanced Value-at-Risk (VaR) methodologies, portfolio risk decomposition techniques and volatility forecasting models.

The analysis evaluates how different quantitative frameworks capture:

- portfolio exposure,
- diversification effects,
- systematic market risk,
- volatility clustering,
- and downside tail-risk behavior under changing market conditions.

The project combines:

- Parametric VaR estimation,
- Asset-Normal portfolio modeling,
- VaR backtesting,
- Component and Relative VaR decomposition,
- Incremental VaR analysis,
- market beta estimation,
- EWMA volatility forecasting,
- and GARCH(1,1) volatility modeling.

The objective is to assess the strengths, limitations and practical implications of different risk management methodologies within a realistic portfolio management environment.

Portfolio Composition

The portfolio was analyzed over the period between November 21, 2022 and March 13, 2025 and consisted of the following Swiss equity positions:

Asset	Position
Nestlé (NESN)	-30,000 shares (short)
Swiss Re (SREN)	+70,000 shares
Novartis (NOVN)	+80,000 shares

The portfolio follows a buy-and-hold strategy in which the number of shares remains constant throughout the investment horizon.

1. Parametric VaR Analysis

The first stage of the analysis focused on the estimation of individual 1-day VaR(99%) measures under the assumption of:

- normally distributed returns,
- constant volatility,
- and historical volatility estimation.

Using 250-day historical volatility estimates, the resulting VaR measures were:

Asset	VaR(99%,1-day)
NESN	CHF 85,323
SREN	CHF 191,707
NOVN	CHF 147,486

The analysis highlights the trade-off between:

- longer estimation windows, which provide more stable volatility estimates,
- and shorter windows, which react more rapidly to market shocks and changing volatility regimes.

Although the parametric approach offers computational efficiency and simplicity, constant volatility assumptions may underestimate tail-risk exposure during periods of elevated market stress.

2. Asset-Normal Portfolio VaR

Portfolio risk was then estimated using an Asset-Normal framework incorporating:

- cross-asset correlations,
- covariance structures,
- and portfolio diversification effects.

Using 250-day volatility and correlation estimates:

Metric	Value
Portfolio VaR	CHF 271,458
Undiversified VaR	CHF 424,516

The results demonstrate significant diversification benefits, with aggregate portfolio risk reduced by approximately 36% relative to the undiversified sum of standalone asset VaRs.

The analysis confirms that correlation structures play a fundamental role in determining total portfolio exposure and overall market risk.

3. VaR Backtesting Analysis

The robustness of the Asset-Normal VaR framework was evaluated through historical backtesting procedures comparing:

- predicted VaR thresholds,
- and realized daily portfolio profit-and-loss outcomes.

Observed exceedance frequencies were:

Sample	Exceedance Rate	Expected
Full Sample	1.60%	1.00%
2024 Sample	1.72%	1.00%

The results indicate that the model slightly underestimated portfolio risk during periods characterized by increased market volatility.

In particular, the higher exceedance rate observed during 2024 suggests that static volatility assumptions may fail to fully capture rapidly changing market dynamics and volatility persistence.

4. Component and Relative VaR Analysis

To better understand portfolio risk decomposition, the analysis incorporated:

- Marginal VaR (MVaR),
- Component VaR (CVaR),
- and Relative Component VaR (RCVaR).

These measures quantify:

- the sensitivity of portfolio VaR to individual positions,
- each asset's contribution to aggregate portfolio risk,
- and the relative importance of each exposure.

Key findings include:

- Swiss Re (SREN) represented the largest contributor to portfolio risk due to its combination of high volatility and large position size.
- Nestlé (NESN), held as a short position, generated negative Relative VaR contributions and therefore acted as a partial hedge during stressed market conditions.
- Novartis (NOVN) displayed the most stable and lowest Relative VaR contribution throughout the sample period.

The analysis demonstrates that portfolio-level risk cannot be evaluated solely through standalone volatility measures, but must also incorporate covariance structures and directional exposure.

5. Incremental VaR and Trade Impact Analysis

A hypothetical portfolio rebalancing scenario was evaluated for March 14, 2025 involving:

- the purchase of CHF 2 million in NESN,
- the purchase of CHF 1 million in SREN,
- and the sale of CHF 1 million in NOVN.

The resulting Asset-Normal VaR estimates were:

Metric	Value
Portfolio VaR Before Trades	CHF 454,128
Portfolio VaR After Trades	CHF 489,436
Incremental VaR	CHF +35,308

Initial linear Incremental VaR estimates overstated the effective increase in portfolio risk because they ignored diversification and covariance interactions.

A full recomputation of the portfolio covariance structure confirmed that the proposed trades generated only a moderate increase in aggregate portfolio exposure.

The results highlight the importance of considering correlation effects when evaluating portfolio reallocations and trade implementation.

6. Systematic Risk and Beta Analysis

The portfolio's systematic market exposure was evaluated through beta estimation relative to the Swiss Market Index (SMI).

Estimated market betas were:

Asset	Beta
NESN	0.757
SREN	1.114
NOVN	0.901

The results confirm:

- Nestlé's defensive characteristics,
- Swiss Re's high sensitivity to broad market movements,
- and Novartis' moderate systematic exposure.

Beta estimates also explain differences in:

- Marginal VaR contributions,
- volatility sensitivity,
- and portfolio reaction to market-wide shocks.

7. EWMA and GARCH Volatility Modeling

To account for time-varying volatility dynamics, the study implemented:

- an EWMA RiskMetrics model with $\lambda = 0.94$,
- and a GARCH(1,1) volatility model.

Estimated conditional volatilities for March 14, 2025 were:

Model	Estimated Volatility
EWMA	0.36%
GARCH(1,1)	0.76%

The comparison highlights important differences between the two methodologies:

- EWMA produces smoother volatility estimates and reacts gradually to market shocks.
- GARCH captures volatility clustering, persistence and mean reversion more effectively.

As a result, the GARCH framework provides a more realistic representation of dynamic market risk under stressed market conditions.

8. Sharpe Diagonal VaR Analysis

The final stage of the project estimated portfolio VaR using a Sharpe Diagonal VaR(99%, 1-day) framework based on forecasted market volatility.

The resulting VaR estimates were:

Model	VaR
EWMA	CHF 41,400
GARCH(1,1)	CHF 276,209
Asset-Normal	CHF 454,128

The EWMA-based VaR materially underestimated portfolio risk due to its smoother volatility adjustment mechanism.

Conversely, the GARCH-based VaR produced estimates materially closer to the Asset-Normal framework than the EWMA-based estimate, confirming its superior ability to capture:

- volatility persistence,
- market clustering,
- and changing risk regimes.

Conclusion

This project demonstrates how different quantitative risk management methodologies produce materially different assessments of portfolio exposure and downside risk.

While simpler models such as:

- Parametric VaR,
- and EWMA volatility estimation,

offer computational simplicity and operational efficiency, they may fail to fully capture:

- volatility clustering,
- correlation instability,
- and tail-risk behavior during stressed market environments.

More advanced approaches such as:

- Asset-Normal portfolio modeling,
- Component VaR decomposition,
- and GARCH volatility forecasting,

provide a more robust framework for measuring dynamic portfolio risk and understanding the interaction between:

- systematic exposure,
- diversification,
- volatility persistence,
- and downside tail-risk.

The analysis further highlights the importance of:

- model validation through backtesting,
- dynamic volatility estimation,
- and covariance-based portfolio analysis

in supporting more accurate risk forecasting and informed portfolio management decisions.